

Daniel Heidorn

| daniel.j.heidorn@gmail.com | danielheidorn.com

Summary

Embedded software engineer with hands-on experience developing controls software for Class 8 trucks. Skilled in embedded software with model-based design (MATLAB/Simulink) and manual coding (C); data science and machine learning (Torch, SKLearn, Seaborn, Pandas); database engineering (relational and non-relational); and project management with methodologies like Agile. Searching for a more senior role with automotive software that offers greater responsibilities and utilizes leadership skills – especially low-level technology like vehicle communications, controls, and BSW interaction.

Education

University of Michigan

Ann Arbor, MI

BACHELOR OF COMPUTER SCIENCE – COLLEGE OF LITERATURE, SCIENCE, AND ARTS

Sep 2018 – May 2022

- Relevant courses include German I & II, Data Structures and Algorithms, Computer Organization, Theory of Computation, Operating System Design, Database Management and Design, User Interface Development, Videogame Development, and Foundations of Artificial Intelligence

Work Experience

Daimler Trucks North America

Detroit, MI

EMBEDDED SOFTWARE ENGINEER

Dec 2023 – Present

- Develop powertrain controls software for heavy-duty vehicle ECUs – especially engine, exhaust and J1939 diagnostics.
- Implement embedded control applications using MATLAB/Simulink for controls and manual coding for diagnostics.
- Validate software functionality across MIL, SIL, HIL, and real-world vehicle testing.
- Perform systems engineering functions to scale customer requirements into software design.
- Create workflow in Jira, integrating systems engineering and agile methodologies to track all tasks.

DATA ENGINEER

Aug 2022 – Nov 2023

- Developed full-stack logistics management software using SQL, Springboot, .NET Core, and Angular.
- Designed and implemented solution to migrate on-premises databases and services to Azure.
- Engineered and implemented various web applications, data structures, ETL pipelines, and analysis.

SHOPFLOOR SYSTEMS INTERN

May 2021 – Aug 2021

- Maintained and updated databases for shop floor MES order systems integrated with MSSQL and SAP S/4HANA.
- Deployed plant floor edge devices, connecting the plant floor operations to a new ERP system.

Personal Projects

J1979onELM – Vehicle Diagnostics Library for Arduino – W.I.P.

SAE J1979 diagnostics library for Arduino to interact with ELM327 UART, providing easy interface to read/diagnose/clear faults, measure instruments and more. Built with message queues and reusability in mind for future UDS project.

TrueValue – Car Value Web Scraper and Analyzer

Data science personal project using Seaborn, Pandas, Matpy, and BeautifulSoup. Scrapes car information from various websites, collecting and analyzing market value, pricing trend, and factor analysis such as mileage affecting market value.

Skills

Programming: C, Matlab, C++, Java (Spring), JS/TS (Nodejs, React, Angular), C# (Unity), Python (Pandas, Seaborn, SKLearn, Keras)

Certifications: Certified Scrum Master; MIT Data Science, Machine Learning, and AI Certified

Affiliations: Chair of EQUAL Detroit-chapter, Society of Automotive Engineers